

MSE 550 Term Paper

Overview

The purpose of this term paper is to demonstrate that you have gained new knowledge in materials characterization. Hopefully, the topic is of interest to you and will help you in your employment or research. All content must be written in clear, grammatically correct formal English.

Dates

Refer to the corresponding Module for specific due dates.

Items

- Tentative Title & First Page with Four Key References listed
- Final Title & Abstract (250 words) & Detailed Outline
- Paper Format and Submission to Canvas
- Reference Format Examples

Specifics

Tentative Title & First Page with Key References

Upload these items as *.pdf file(s)

- Title can start with “The Use of [technique] for”
- Topics should be limited to topics relevant to Materials Characterization.
- First page with your four Key References. They are to come from archival refereed journal articles (less than four years old).
- The following items cannot be used as one of your Key References - textbook, proceeding articles, trade articles, or internet links.

Final Title & Abstract & Detailed Outline

Upload these items as *.pdf file(s)

Final title of your paper

Abstract of your paper

Detailed Outline:

- section headings (Introduction, Description, Case Study I and Case Study II, and Summary)
- sub-headings in each section

Paper Format and Submission to Canvas

Upload paper as a single *.pdf file

It is extremely important to submit your paper by the due date to facilitate proper Peer-Review assignments.

Late submission will result in a grade reduction.

- Approximately 3000 words of text
(Note: Title page, Table of Contents, References, Figure pages are not included in the count.)
- 12 point Times New Roman font, single spacing, 1" margins all round
- Order of Pages
 - Title, Abstract, Name, Course #, Due date (all on the first page)
 - Table of Contents (new page)
 - Introduction (new page)
 - Fundamental description of the technique - demonstrate that you understand the fundamental concepts.
 - Case Study I
 - Case Study II
 - Summary
 - References (numbered at first use, see format below)
- Figures can be either placed within the text of your paper, or placed at the end of the paper.
 - – Figures (numbered at first use and must include figure captions)

Example Reference Formats:

Article:

- 'Optimization of the Zinc Oxide Electron Transport Layer in Organic Solar Cells by Annealing and Yttrium Doping,' S. Das and T.L. Alford, Royal Society of Chemistry 5, 45586-91 (2015).
- 'Computational Modeling of Optical Properties in Aluminum Nanolayers Inserted in ZnO for Solar Cell Electrodes,' E.R. Rwenyagila, B. Agyei-Tuffour, K.O. Onogu, O. Akin-Ojo, M.G.Z. Kana, T.L. Alford, and W.O. Soboyejo, Optic Letters 40, 3914-17(2015).

Book:

- Fundamentals of Nanoscale Film Analysis, T.L. Alford, L.C. Feldman, and J.W. Mayer, Springer 2007, page 73- 75.
- ASM Metals Handbook 9th ed. Vol. 10 Materials Characterization, ASM International 1986, pp 2310-2315.

Internet:

- <https://www.phi.com/surface-analysis-applications.html>, (accessed 07 Feb 2020).

Lecture Notes:

- MSE 550 Lecture Notes #5, T.L. Alford and N.D. Theodore, Arizona State University (2016).

Patent:

- U.S. Patent 9147605, Thin Film Devices and Low Temperature Process to Make Thin Film Devices, S. Das and T.L. Alford (2015).
- U.S. Patent Application 20160027955 A1, Microwave-Annealed Indium Gallium Zinc Oxide Films and Methods of Making the Same, T.L. Alford and A. Dhar (2016).

Improper Citation

- Not referencing a text or figure properly
- Typo in citations
- Quoting text verbatim without due reference will be considered plagiarism